AI-Driven Smart Contracts on Blockchain

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Enhancing Development, Implementation, and Flexibility with Artificial Intelligence

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ABSTRACT

This research investigates how Artificial Intelligence (AI) enhances smart contract development, implementation, and flexibility on blockchain platforms. It highlights AI's adaptive automation and identifies a benefit-risk paradox between efficiency and security.

INTRODUCTION

Smart contracts automate agreements but lack flexibility. Integrating AI enables adaptive logic, predictive automation, and smarter execution.

PROBLEM STATEMENT

Smart contracts face platform limitations, rigidity, and security issues. AI integration enhances flexibility, automation, and intelligence.

AIM & OBJECTIVES

Aim: Explore AI-driven solutions enhancing smart contracts. Objectives: Investigate flexibility, automation, risk-benefit balance, and platform challenges.

METHODOLOGY

Design: Technical Action Research (TAR). Quantitative survey of 53 blockchain/AI professionals. Tools: SPSS v29, statistical and cross-tab analyses.

KEY FINDINGS

AI improves flexibility (94.3%) and automation (86.7%). Top benefits: efficiency (96.2%) and analytics (88.7%). Main risks: skill gaps (94.3%) and compliance (90.6%).

DISCUSSION

AI brings flexibility but adds complexity. Ethereum's dominance shows maturity; Hyperledger's difficulty reflects enterprise complexity. Benefit-risk paradox remains strong.

CONCLUSION & RECOMMENDATIONS

AI boosts flexibility and automation but requires stronger security and skill frameworks. Developers should adopt AI responsibly, aligned with SDG 9.